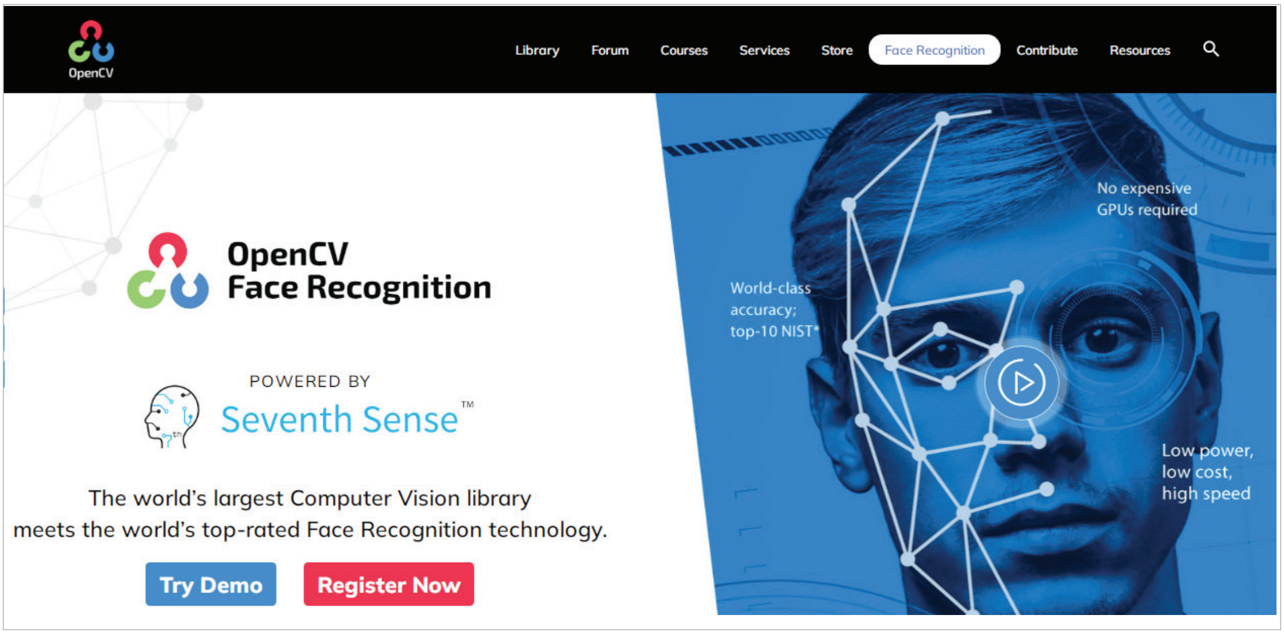

Figure 1: OpenCV library for computer vision

IoT meets computer vision for a wide range of applications

The integration of AI and Internet of Things (IoT) has opened up new avenues for the application of computer vision in a range of industries and use cases.

Intelligent spaces: One of the primary applications of computer vision in the context of AI and IoT is in the field of smart homes and buildings. With the integration of sensors and cameras, buildings can be equipped with systems that monitor and analyse data such as temperature, light, and occupancy. This information can be used to automate various processes such as adjusting the temperature or turning off lights when a room is unoccupied.

Healthcare: By integrating cameras and sensors into medical equipment and devices, healthcare providers can collect data and images to aid in diagnoses and treatment. For example, computer vision algorithms can analyse X-ray or MRI scans to detect early signs of disease or abnormalities.

Manufacturing: In the manufacturing industry, computer vision and AI can be used to optimise production processes and improve quality control. By integrating cameras and sensors into manufacturing equipment, manufacturers can collect data and images to identify defects and improve efficiency. Additionally, computer vision can be used to monitor employee performance and ensure compliance with safety regulations.

Defence industry: Computer vision, AI and IoT have also found several applications in the defence industry.

- *Surveillance and security:* Computer vision algorithms can be integrated with cameras and sensors to monitor sensitive areas and detect potential threats. The system can identify individuals and objects, and track their movements in real-time, enabling a quick response to any security breaches.
- *Autonomous systems:* Drones and unmanned vehicles equipped with computer vision and AI can be used for surveillance and reconnaissance, as well as for carrying out missions such as search and rescue operations or delivering supplies to remote areas.
- *Predictive maintenance:* In the defence industry, equipment downtime can have serious consequences. By integrating sensors with computer vision and AI, maintenance teams can monitor equipment performance in real-time and predict when maintenance is required, reducing downtime and ensuring readiness.

Agriculture: Computer vision can be used to monitor crop growth, detect plant diseases or pests, and optimise irrigation and fertilisation. This can improve crop yield and reduce waste.

Retail: Retailers can use computer vision and AI to analyse customer behaviour, such as how they interact with products and which products they prefer. This data can be used to improve store layout, inventory management, and personalised marketing strategies.

Traffic management: Integrating sensors and cameras with computer vision and AI can help monitor traffic flow, identify traffic patterns, and optimise traffic lights. This can reduce traffic congestion and improve road safety.